

Early birds

Birds evolved from dromaeosaur-like dinosaurs during the Jurassic Period. The first birds had skeletons like those of *Microaptor* (page 198). Over time, as birds adapted to life in the air, they evolved huge flight muscles and lost their teeth, tails, and claws, making them more lightweight.



FAMILY FACT FILE

Key features of modern birds

- Feathered body and wings
- Toothless beaks
- Tail bones fused into a stump
- No finger claws or small finger claws
- Deep keel bone on breast to anchor large flight muscles
- Semicircular wrist bone to aid flapping

When

Birds first appeared in the Late Jurassic and have been in the skies ever since.



Modern birds have a toothless beak, but *Archaeopteryx* had jaws and teeth typical of a carnivorous dinosaur.

Very long feathered arms with flight feathers

Archaeopteryx

Confuciusornis

con-FEW-shus-OR-niss

- **When** 130–120 million years ago (Early Cretaceous)
- **Fossil location** China
- **Habitat** Woodlands of Asia
- **Length** 1 ft (0.3 m)
- **Diet** Probably seeds

Confuciusornis was the earliest toothless bird and the first known to have a beak. It also had a stumpy tail like that of modern birds, but it lacked strong flight muscles. Thousands of fossils of *Confuciusornis* have been found in China, and some of the adults have very long tail feathers. These may be male ornaments that were displayed to attract females during courtship.



Archaeopteryx

ar-kee-OP-ter-ix

- **When** 150 million years ago (Late Jurassic)
- **Fossil location** Germany
- **Habitat** Forests and lakes of western Europe
- **Length** 1 ft (0.3 m)
- **Diet** Insects, probably reptiles

When the first complete fossil of *Archaeopteryx* was discovered in 1861, scientists were amazed—it looked like a cross between a



dinosaur and a bird. It had a fully feathered tail and wings, yet it also had dromaeosaur-like claws on its hands, bones along its tail, and jaws with teeth instead of a beak. *Archaeopteryx* is the oldest known member of the bird family. It was the size a pigeon and had long flight feathers, but it lacked the powerful muscles needed for flapping flight and was probably more of a glider than a flapper.

◀ **THIS FOSSIL** of *Archaeopteryx* was discovered in Germany. Amazingly clear impressions of feathers on the arms and tail are preserved in fine-grained limestone.